Goursat’s Theorem

> **Theorem 0.0.3.** Let G be an open set and let $f : G \to \mathbb{C}$ be a holomorphic function.
> Then,
>
> $$\int_T f = 0$$
>
> for every triangular path T in G.

Proof. Let $T_0 = [a, b, c, a]$ be the triangular path in G, and let $d = \operatorname{diam} T$ and $L = \operatorname{length} T$.
And, construct four subtriangles

$$T^{(1)} = \left[a, \frac{a+b}{2}, \frac{a+c}{2}, a\right],\ T^{(2)} = \left[\frac{a+b}{2}, b, \frac{b+c}{2}, \frac{a+b}{2}\right],\ T^{(3)} = \left[\frac{a+c}{2}, \frac{b+c}{2}, c, \frac{a+c}{2}\right],\ T^{(4)} = \left[\frac{a+b}{2}, \frac{b+c}{2}, \frac{a+c}{2}, \frac{a+b}{2}\right]$$

Then,

$$\int_{T_0} f = \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f$$

Thus, by the triangle inequailty,

$$\left|\int_{T_0} f\right| \leq \left|\int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f\right| \leq 4 \cdot \max_{i=1,2,3,4} \left|\int_{T_i} f\right|$$

Let $T_1 = T^j$, where the j maximizes the above integral. Apply the same process for $T_0, T_1, \ldots, T_{n-1}$, we obtain that a sequence of nested compact sets $\{\mathcal{T}_n\}$, where $\mathcal{T}_n$ is the convex hull of T_n. So, the intersection

$$\bigcap_{n=0}^{\infty} \mathcal{T}_n = \{z_0\}$$

is a singleton. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies \left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon$$

since f is holomorphic. Moreover, by the construction, there exists $N \in \mathbb{N}$ such that

$$T_N \subseteq B(z_0; \delta)$$

Now, the triangle inequality again,

$$\left|\int_{T_N} f(z)dz\right| = \left|\int_{T_N} f(z) - f(z_0) - f'(z_0)(z - z_0)dz\right| \leq \varepsilon \int_{T_N} |z - z_0||dz| = \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N$$

Finally, since $\operatorname{diam} T_N = 2^{-n} \operatorname{diam} T_0$ and $\operatorname{length} T_N = 2^{-n} \operatorname{length} T_0$,

$$\left|\int_{T_0} f(z)dz\right| \leq 4^N \cdot \left|\int_{T_N} f(z)dz\right| \leq 4^N \cdot \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N \leq \varepsilon \cdot \operatorname{diam} T_0 \cdot \operatorname{length} T_0 \to 0 \quad \text{as} \quad \varepsilon \to 0$$

$\square$

「Thm. Let G be an open set of $\mathbb{C}$, and T be a triangluler path contained in G. For a point a inside in T, suppose that f is Holomorphic on $G \setminus \{a\}$. If f is bounded on for some neighborhood of a, then

$$\int_T f = 0.$$

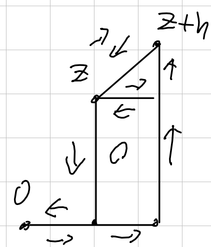
Proof. For fixed a, construct a subtriangle T^1, T^2, T^3, T^4 from T. Then, for some T^k, $a \in T^k$. (There are at most three case). Put this $T_1 = T^k$. Then, inductively, we can construct $\{T_n\}$ compact nested, and

$$\left| \int_T f \right| \le \left| \sum_{a \in T_n^k} \int_{T_n^k} f \right| \le \frac{3L}{2^n} \cdot \sup |f| \to 0 \text{ as } n \to 0.$$

Thm. A holomorphic function $f: D \to \mathbb{C}$ has a primitive where D is open disk.

Proof. WLOG, let $D = B(0; R)$. Given $z = a + bi \in D$, define

$$F(z) = \int_{\gamma_z} f(s)\, ds \qquad \text{where } \gamma_z = [(0,0),(a,0),(a,b)]$$



Now, let $h \in \mathbb{C}$ with $0 < |h| < R - |z|$. Then,

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(s)\, ds - \int_{\gamma_z} f(s)\, ds = \int_{[z, z+h]} f(s)\, ds$$

using Goursat Thm in Here, with holomorphic.

Given $\varepsilon > 0$, there is $\delta > 0$ such that

$$|z - s| < \delta \Rightarrow |f(z) - f(s)| < \varepsilon.$$

So,

$$\left| \frac{1}{h} \cdot \int_{[z,z+h]} f(s)\, ds - f(z) \right| = \left| \frac{1}{h} \cdot \int_{[z,z+h]} f(s) - f(z)\, ds \right| < \varepsilon, \text{ for small enough } h.$$

∎

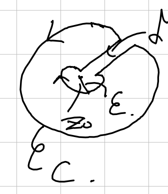
Corollary. If f is Homolophic on an open disk, then for any closed curve γ in the disk,

$$\int_\gamma f = 0.$$

┌Cauchy's Integral Formula, Revised.

┌Thm. Let $f: G \to \mathbb{C}$ be a Holomorphic on open $G \subset \mathbb{C}$.
If $z_0 \in G$ and $\overline{B_r(z_0)} \subseteq G$, then for a Curve $C: z_0 + re^{it}$ $(0 \le t \le 2\pi)$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \oint_C \frac{f(z)}{z - z_0} \, dz .$$



Proof. Denote $D = B_r(z_0)$. Let $z_0 \in D$ be fixed, and Consider a Curve $\Gamma_{\delta,\varepsilon}$ as
Since $\dfrac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, $\displaystyle\int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0} \, dz = 0$.

Taking $\delta \to 0$, then we obtain

$$\lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0} \, dz = -\int_{C_\varepsilon} \frac{f(z)}{z - z_0} \, dz + \int_C \frac{f(z)}{z - z_0} \, dz = 0.$$

Meanwhile, $\dfrac{f(z)}{z - z_0} = \dfrac{f(z) - f(z_0)}{z - z_0} + \dfrac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\displaystyle\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z) - f(z_0)}{z - z_0} \, dz = 0$.

Now,
$$\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z_0)}{z - z_0} \, dz = f(z_0) \cdot \lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{1}{z - z_0} \, dz$$
$$= f(z_0) \cdot \lim_{\varepsilon \to 0} \left(\int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it} \, dt \right)$$
$$= f(z_0) \cdot 2\pi i .$$

$$\int_C \frac{f(z)}{z - z_0} \, dz = 2\pi i \, f(z_0).$$

Consequently,

$$\int_C \frac{f(z)}{z - z_0} \, dz = \lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0} \, dz + \int_{C_\varepsilon} \frac{f(z)}{z - z_0} \, dz = 2\pi i \cdot f(z_0).$$

┌Corollary.
$$\int_C \frac{f(z)}{(z - z_0)^{n+1}} \, dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}$$

Thm. Suppose f is Holomorphic on an open G containing a closed disk $\underset{D}{\wedge}$ having center z_0,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z-z_0)^n \qquad (z \in D)$$

Proof.

$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} \, d\mu \qquad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z-z_0)^n \, d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} \, d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n.$$

Corollary, Liouville's Theorem.

If $f: \mathbb{C} \to \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \le \frac{1}{R} \cdot \sup_{C_R} |f| \to 0$ as $R \to \infty$. Therefore, $f' = 0$

Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.

observe that $\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$

The terms $a_{n-k} \cdot z^{-k}$ $(k = 1, \cdots, n)$ converges to zero as $|z| \to \infty$, thus for some $R > 0$,

$$|P(z)| \ge 2|a_n| \cdot z^n \quad (|z| > R)$$
$$\ge 2|a_n| \cdot R^n$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \le R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.